

SECJ2154 (OOP)

PROJECT REPORT

Mini Project Title:

Lost and Found Reporting System

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MINI PROJECT DESCRIPTION

1. Introduction

The Lost and Found Reporting System aims to assist students and campus staff in efficiently reporting lost or found items. It offers a centralized platform where users can log reports, monitor item statuses, and manage records of found property. This system helps reduce confusion and improve the chances of recovering lost items through organized digital tracking.

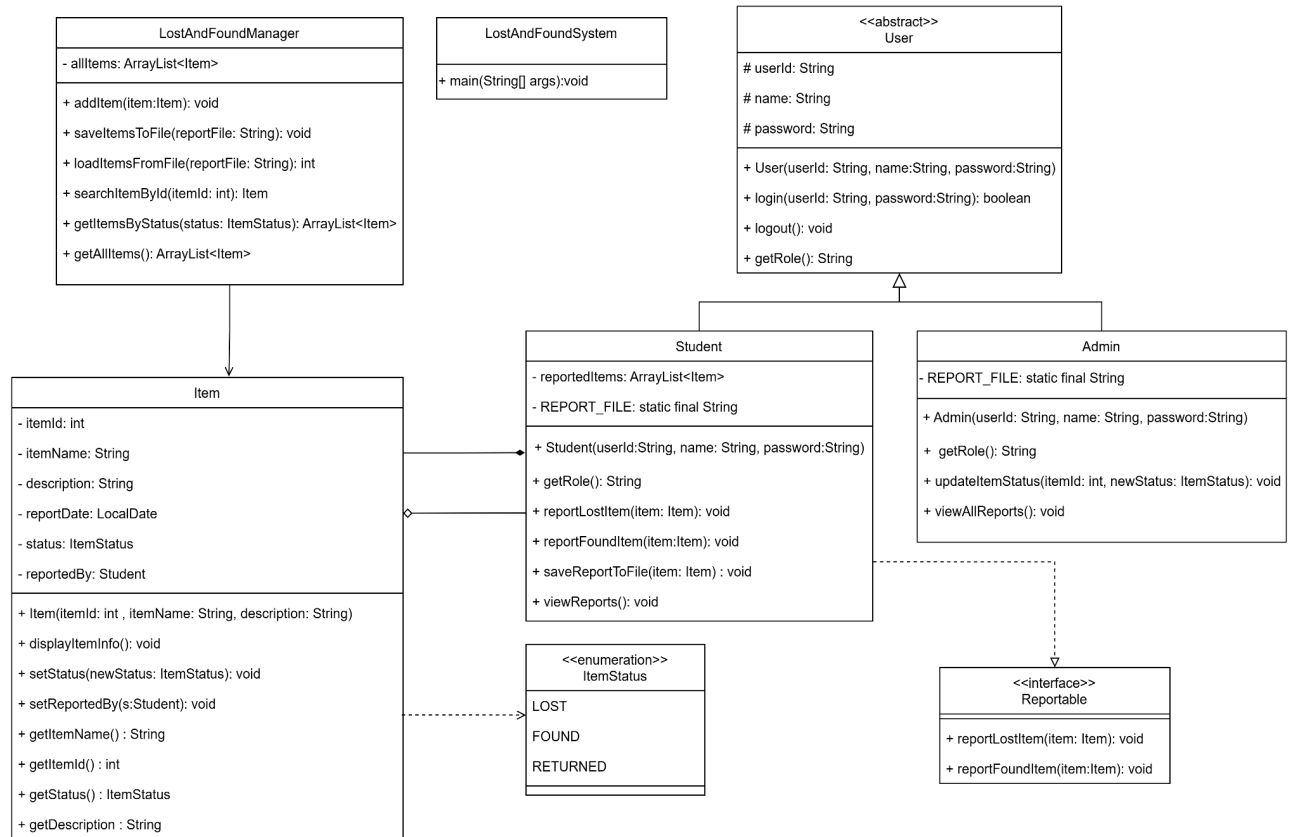
2. Problem statement

Currently, the reporting and tracking of lost or found items is primarily carried out in a manual or informal manner, often relying on notice boards or social media. This approach results in challenges such as tracking difficulties, duplicate reports, and miscommunication. To address these issues, we suggest implementing a digital system that enables users to report lost or found items, manage their statuses, and maintain a structured record of all reports.

3. Objectives

1. To develop a centralized system rather than record manually that allows students and staff to report lost or found items in a structured and accessible manner.
2. To make sure students understand and could implement key object-oriented programming principles such as inheritance, abstraction and polymorphism in real-world applications.
3. To ensure data persistence by incorporating file handling for storing and retrieving item and user information.
4. To provide a user-friendly interface that facilitates all the functionalities should have in this system.

4. Project Design (Class Diagram)



Person-in-charge	Classess
Chau Ying Jia	1. Student 2. LostAndFoundManager
Nawwarah Auni	1. Admin 2. Item 3. Reportable
Mohamed Ali	1. User 2. ItemStatus

5. Project Output (Screenshots)

i) Student

1. Run the program

```
PS C:\JavaPrograms\OOP\miniProject> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\chaury\AppData\Roaming\Code\User\workspaceStorage\ab76fb4728f55d0126e02a401e1913ff\redhat.java\jdt_ws\miniProject_b002f11d\bin' 'LostAndFoundSystem'  
Please enter your ID: █
```

2. Enter ID and password

```
Please enter your ID: S001  
Please enter your password: 1234  
Login successful! Welcome Ali (Student)
```

MENU

1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout

Choose an option (1/2/3/4): █

Login successful as a student.

3. Choose selection '1' to report lost items.

MENU

1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout

Choose an option (1/2/3/4): 1

Enter item name: █

If user enter input other than 1,2,3 or 4, it will throw an error message (throw an exception)

4. Enter lost item details

```
Choose an option (1/2/3/4): 1
Enter item name: wallet
Enter item description: black colour CK brand lost in N28
Lost item reported: wallet
```

“Lost item reported: wallet” is an automated output

If a student leaves the input blank, an exception error output will be thrown:

```
MENU
1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout
Choose an option (1/2/3/4): 1
Enter item name:
Error: Item name cannot be blank!
```

```
MENU
1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout
Choose an option (1/2/3/4): █
```

5. Item will be saved into report file called stuReportFile.txt

```
≡ stuReportFile.txt
1  ItemID  Name      Status  ReportedBy  Description
2  -----
3  1      wallet    LOST    Ali         black colour CK brand lost in N28
4
```

6. Choose selection '2' to report found item

```
MENU
1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout
Choose an option (1/2/3/4): 2
Enter item name: █
```

If user enter input other than 1,2,3 or 4, it will throw an error message (throw an exception)

7. Enter found item details

Choose an option (1/2/3/4): 2
Enter item name: wallet
Enter item description: call 0175656565
Found item reported: wallet

8. Item will be saved into report file called stuReportFile.txt

≡ stuReportFile.txt

1	ItemID	Name	Status	ReportedBy	Description
2	-----				
3	1	wallet	LOST	Ali	black colour CK brand lost in N28
4	2	wallet	FOUND	Sara	call 0175656565
5					

9. Choose '3' to view all reports.

MENU

1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout

Choose an option (1/2/3/4): 3

10. The report load from stuReportFile.txt to let students view the report

--- All Reported Lost & Found Items ---

ItemID	Name	Status	ReportedBy	Description

1	wallet	LOST	Ali	black colour CK brand lost in N28
2	wallet	FOUND	Sara	call 0175656565

11. Students can logout by pressing '4' after completion.

MENU

1. Report Lost Item
2. Report Found Item
3. View All Reports
4. Logout

Choose an option (1/2/3/4): 4

Ali has logged out.

ii) Admin

1. Enter ID and Password

```
Please enter your ID: A001
Please enter your password: adminpass
Login successful! Welcome Admin (Admin)
```

MENU

```
1. View all items
2. Update item status
3. Logout
Choose an option (1/2/3): █
```

Login successful as an admin.

2. Enter '1' to view all items.

MENU

```
1. View all items
2. Update item status
3. Logout
Choose an option (1/2/3): 1
```

--- All Lost & Found Items ---

ItemID	Name	Status	ReportedBy	Description
1	Wallet	LOST	Ali	Found at the balcony
2	Scissors	FOUND	Ali	Found from Ali
3	Wallet	RETURNED	Ali	Lost at the countertop
4	Watch	RETURNED	Ali	Lost at the canteen

All reported items displayed.

If the user enters input other than 1,2, or 3, it will throw an error message (throw an exception).

3. Enter '2' to update item status.

MENU

```
1. View all items
2. Update item status
3. Logout
Choose an option (1/2/3): 2
```

4. Enter the item ID to update the item status.

```
Enter item ID to update: 2
```


5. Enter the new status.

```
Enter new status (LOST/FOUND/RETURNED):  
returned  
Item status updated: Scissors -> RETURNED
```

The status entered will automatically be converted to uppercase.

6. Admin can enter '3' to log out.

```
MENU  
1. View all items  
2. Update item status  
3. Logout  
Choose an option (1/2/3): 3  
Admin has logged out.
```

7. The report load from stuReportFile.txt to let admin view the report

ItemID	Name	Status	ReportedBy	Description
1	Wallet	LOST	Ali	Found at the balcony
2	Scissors	RETURNED	Ali	Found from Ali
3	Wallet	RETURNED	Ali	Lost at the countertop
4	Watch	RETURNED	Ali	Lost at the canteen

6. Conclusion

In conclusion, this project was created to offer a better way of reporting and tracking lost and found items by replacing the manual method with a structured digital system. It uses object-oriented programming (OOP) principles such as inheritance, association, abstraction, interface, and polymorphism to build clear and manageable code. The system also uses file handling to store and retrieve the report and includes a simple and user-friendly interface. The output shows the main functions of the system, including how users interact with it and how reports are managed. The successful use of class diagrams, system features, and visual outputs supports the main goal of the project. This proves that the system can improve the process of managing lost and found items in an organized and effective way.

7. Appendices

Link for codes:

https://drive.google.com/drive/folders/1KYwypJnRZ8rncARHRWSPCgxAgx_SWU_G?usp=sharing